Ophthalmic Epidemiology, 19(5), 306–316, 2012
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ISSN: 0928-6586 print/1744-5086 online
DOI: 10.3109/09286586.2012.712192

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ORIGINAL ARTICLE

Andhra Pradesh Eye Disease Study – Visual Function Questionnaire: Further Improvements in Psychometric Properties using Rasch Analysis

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ABSTRACT

Purpose: Previous Rasch analysis of the Andhra Pradesh Eye Disease Study-Visual Function Questionnaire (APEDS-VFQ) lacked comprehensiveness, specifically, dimensionality (whether it measures single/multiple constructs). Therefore, using the Rasch model this study provides a detailed assessment of psychometric properties of the APEDS-VFQ.

Methods: A total of 351 visually impaired adults (mean age, 43.3 years) were verbally administered the APEDS-VFQ. Rasch analysis was used to assess the psychometric properties of the instrument.

Results: Participants could distinguish only three categories of difficulty, so response categories were reduced from five to three. A single item ("reading small prints in newspaper/magazines", infit mean square 1.54) misfit the model. The overall pattern of fit statistics for item and person measures suggested that the underlying construct (visual ability) is not unidimensional. When the items were grouped into subsets based on functional requirements (resolution, contrast sensitivity, illumination and peripheral vision) and separate person measures were estimated for each of these domains, the first principal component contained the visual ability and accounted for 72% of the variance. Item measure distributions could be divided into 18 strata, and item-separation reliability was 0.99. Person measures could be divided into three statistically distinct strata and the person-separation reliability was 0.81.

Conclusions: The APEDS-VFQ is a precise measure of visual ability in visually impaired adults in India. Similar to other visual function questionnaires developed for the Western population, our results demonstrate that visual ability is a two-factor composite latent variable; one dimension heavily influences reading and the other most heavily influences peripheral vision (mobility).

KEYWORDS: Rasch analysis, Andhra Pradesh Eye Disease Study-Visual Function Questionnaire, Psychometric properties, India

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INTRODUCTION

The Andhra Pradesh Eye Disease Study-Visual Function Questionnaire (APEDS-VFQ) was developed to assess the level of difficulty of various aspects of visual function in an Indian adult low vision population.[1] Previous Rasch analysis of the APEDS-VFQ reported adequate measurement precision and targeting of the instrument in 135 adults with low vision.[2] However, few misfitting items were reported and shortening of the rating scale was suggested. Although procedures for re-engineering existing instruments (for example, deleting misfitting items) have been recommended, the potential of an optimized version of the APEDS-VFQ was not explored. Considered together, these findings suggest that the APEDS-VFQ in its native form is likely suboptimal. Furthermore, certain components of the APEDS-VFQ have not been investigated either in detail or completely. These include: (1) dimensionality (the extent to which all items are measuring the same underlying construct, ie unidimensionality, or if they are also measuring a different construct, in which

Received 30 May 2011; revised 31 December 2011; accepted 06 March 2012

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case the instrument lacks unidimensionality),[3,4] that has not been assessed in detail; and (2) differential item functioning (DIF, which tests whether items perform consistently across subgroups of the population), which has yet to be investigated.

Unidimensionality constitutes an important attribute of a good measure.[3] This property is an important prerequisite to generate a valid total score for an instrument and the user can be certain that a single underlying construct is being measured.[5–7] Although previous Rasch analysis of the APEDS-VFQ used fit statistics to investigate dimensionality, this constitutes a limited aspect of the assessment. Recent studies have indicated that fit statistics alone are an insufficient indicator of unidimensionality; a thorough assessment using principal components analysis (PCA) of residuals is required.[8] DIF is another important related property of a good measure as its presence violates the assumption of unidimensionality.[9] DIF occurs when an item performs differently for one subgroup compared to another, despite being equal on the amount of the construct in both subgroups.[10] Thus, DIF analysis can help to determine if items are appropriate for use in various subgroups in a heterogeneous sample.

Using some of the widely used VFQs such as the Activities of Daily Vision Scale, National Eye Institute Visual Function Questionnaire, Visual Functioning Index-14, Visual Activities Questionnaire and the Activity Inventory, Massof demonstrated that visual ability is a composite variable with two factors; one most heavily influences reading function and the other most heavily influences mobility function.[11–14] Given this finding, the aim of the present study was to examine whether the underlying construct (visual ability) of the APEDS-VFQ exhibits a similar pattern. A secondary aim was to investigate the behavior of the response categories and in case of deficiencies, to propose a revised rating scale. Finally, Excel sheets that convert raw to Rasch-scaled scores for those who wish to use the benefits of interval-level measurement are provided.

METHODS

Andhra Pradesh Eye Disease Study-Visual Function Questionnaire

The APEDS-VFQ contains 16 items (Table 1). Participants rated the level of difficulty on a five category scale in performing day-to-day tasks. The response options range from 0 ("no difficulty") to 4 ("cannot manage"). Intermediate response options include 1 ("little difficulty"), 2 ("moderate difficulty") and 3 ("great deal of difficulty but can still manage"). The instrument was scored as recommended by the developers. A higher APEDS-VFQ score represents greater difficulty with visual function.

TABLE 1 Item content of the revised 16-item Andhra Pradesh Eye Disease Study-Visual Function Questionnaire.

Original item no.	Description
1.	Reading small prints in the newspaper/magazines
2.	Recognizing small objects (for example, inserting a thread in a needle)
3.	Recognizing people near you
4.	Recognizing people across the street
5.	Recognizing the bus number
6.	Seeing objects in bright light due to glare
7.	Seeing objects in poorly lit surroundings?
8.	Adjusting from darkness to bright light
9.	Adjusting from bright light to darkness
10.	Identifying colors
11.	Recognizing traffic signals/lights
12.	Noticing objects off to the side, when you are walking and looking straight ahead
13.	Noticing objects on the floor near you, when you are walking around the house and looking straight ahead
14.	Reaching for an object, because it is farther or closer than you thought (eg, to take a glass)
15.	Climbing up or down steps because they were closer or further away than you thought
16.	Estimating the distance of a vehicle (a bus coming towards you) while crossing the road

The framing question for all items is "How much difficulty do you have in…"

Subjects

Participants were visually impaired adults referred to the Vision Rehabilitation Centres, LV Prasad Eye Institute, India. The APEDS-VFQ was administered by trained interviewers in a face-to-face interview to each participant prior to any examination.

Participants were aged 18 years or older, had visual impairment (best-corrected visual acuity in the better eye <20/60 or ≥20/60 with restricted visual fields) and could respond to the instrument. Ethical approval was obtained from the LV Prasad Eye Institute Ethics Committee and all participants provided informed consent. The study was conducted in accordance with the tenets of the Declaration of Helsinki.

Sociodemographic data was extracted from clinical records. Participant characteristics of those who provided responses to the APEDS-VFQ are shown in Table 2.

Procedures

Each participant underwent assessment for comprehensive low vision rehabilitation; procedures have been described in detail elsewhere.[15] Specifically, distance visual acuity (habitual and best-spectacle corrected) was recorded using a Bailey-Lovie logarithm of the minimum angle of resolution (logMAR) chart.[16] All assessments were performed monocularly and binocularly.

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TABLE 2 Sociodemographic and clinical characteristics of the 351 participants who completed the 16-item Andhra Pradesh Eye Disease Study-Visual Function Questionnaire.

Characteristic		n (%) or mean ± SD
Age (years)		43.3 ± 17.8
Sex		
	Male	269 (77)
	Female	82 (23)
Visual acuity in better eye		
	≥20/60*	60 (17)
	<20/60–20/200	205 (58)
	<20/200	86 (25)
Mean visual acuity in better eye	LogMAR (Snellen)	0.83 ± 0.42 (6/38-2)
	Range	0.0 to 3.00 (20/20 to light perception)
Education level$		
	Illiterate	27 (8)
	Up to 9 years	88 (25)
	10–12 years	91 (26)
	> 12 years	127 (36)
Duration of vision loss (years)		
Median		5
Systemic co-morbidity#		
Present		110 (31)
Absent		239 (68)

*These patients had restricted visual fields (less than 20 degrees in better eye).
$Data was missing for 18 cases.
#Includes diabetes, hypertension, asthma, coronary artery disease. Data was missing for two cases.
SD, standard deviation; LogMAR, logarithm of the minimum angle of resolution.

Rasch Analysis

The matrix of responses from 351 participants was subjected to Rasch analysis using the Andrich rating scale model for polytomous data.[17] The theory behind the Rasch measurement model has been described elegantly by Massof.[18,19] Rasch analysis is an iterative procedure that estimates interval measurement from ordinal data and the unit is logits (log-odd units).[19]

Winsteps software[20] (version 3.68) was used to conduct the Rasch analysis, and estimates of person ability and item difficulties were obtained. There are a series of components to Rasch analysis that have been described elsewhere in detail.[21,22] Briefly, we assessed the properties as described below.

Unidimensionality was assessed by examining the item fit statistic, infit mean square (MnSq), which has an expected value of 1.0 and is preferable for assessing item fit as it is less sensitive to sample size compared to standardized *z* values.[24] We used infit MnSq values between 0.7 and 1.3 to indicate item fit.[21] Items with MnSq values less than 0.7 suggest presence of redundant items and values higher than 1.3 indicate unusual and/or inappropriate response patterns, and signal misfit. While some researchers believe that misfitting items should be deleted to optimize the instrument's properties, we did not pursue this instantaneously, as we believe that item misfit *per se* should not be a cause for removing items. Furthermore an instrument is considered to have acceptable unidimensionality when at least 95% of the items fit the Rasch measurement model.[25] The presence of misfitting items should rather alert the researcher that there might be a problem requiring further investigation (such as DIF analysis – described later). Item misfit can be sample-dependent, so we investigated fit statistics for person measures, at times removing grossly misfitting persons (standardized infit, z >2.5) who distorted the measurement system which could potentially fix the item misfits. Some of the reasons for person misfit could be carelessness, fatigue, boredom or disinterest and this could threaten the instrument's validity. Closer analysis of these response patterns is required before a sound rationale can be established for removing all misfitting persons. However, it is generally accepted that 2.5% of the sample, by chance, may not demonstrate acceptable fit.[12]

Subsequently, we investigated the person measure fit statistics and identified the outliers. Then we grouped participants by their fit statistics to test the hypothesis that misfitting persons represent distortion by identifiable confounding variables in the sample. We expected two possible outcomes: (1) non-uniform DIF (ie members of one group are favored up to a level on the ability scale, and from that point on, the relationship is reversed[26]) will vary across infit groups (steepest item characteristic curves for small MnSq values and shallowest item characteristic curves for large MnSq values); and/or (2) uniform DIF (when a group performs better than another group on all ability levels[26]) will occur for groups of misfitting persons. The items that showed the most MnSq-dependent non-uniform and/or uniform DIF would be considered the strongest candidates for being edited from the instrument because they are sensitive to contamination by confounding variables in the person sample. Furthermore, the presence of DIF suggests that the item is measuring an additional construct, or dimension that may not be relevant to the intended construct. DIF was considered notable if >1.0 logits.[21]

In addition to DIF, we also investigated differential person functioning that occurs when a person may function differently on two groups of items after controlling for an overall measure.[27] For this analysis, we divided the 16 items into 4 groups based on functional requirements: (1) those dependent on resolution; (2) those dependent on illumination; (3) those dependent on contrast sensitivity; and (4) those dependent on peripheral vision. Subsequently, we estimated the person measures from each subset of items (with item measures and structure calibrations anchored to their originally estimated values) and then performed exploratory factor analysis (using varimax rotation) on the set of person measures plus logMAR acuity, restricting the analysis

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to two factors. This analysis revealed if there was a two composite latent variable in the underlying construct.

Principal components analysis of the item measure residuals was undertaken to further assess the dimensionality of the instrument. We used the following two criteria to establish unidimensionality: (1) a cut-off variance of 60% accounted for by the first factor (ie dimension); and (2) the eigenvalue of the first contrast should be smaller than 3.0.[8]

Although previous Rasch analysis of the APEDS-VFQ suggested shortening of the rating scale by combining the two end categories, the resultant new category was, however, not labeled. Furthermore, the rater use of the revised categories in a new sample has not been evaluated. Thus, we investigated the behavior of the original rating scale in our population. Specifically, we assessed the response format for location of the thresholds using category probability curves (CPC). The threshold is the point of intersection between two adjacent categories where the response to either of the two adjacent categories is equally probable.[23] The thresholds are expected to increase monotonically. For a well-fitting model, one would expect an ordered set of response thresholds (first followed by second threshold, and so on) represented by a range of hills arranged sequentially in the CPC. However, if threshold disordering (for example, the third threshold is located between the first and second) occurs, then categories need to be combined to repair the disordering.

Reliability of the estimated measures was determined by examination of the person and item separation indices and their related separation reliability. Separation (G) is defined as the ratio of the true spread of the measures with their measurement error, and refers to the number of distinct strata that can be identified in the distribution of participants and items.[23,25] Consequently the item separation index provides an estimate (in standard error, SE, units) of the spread or separation of items along the measurement construct, and the person separation index gives an estimate of the spread or separation of participants along the measurement construct. A separation of 2.0 is considered good and enables the distinction of three groups or strata (number of distinct strata = (4G+1)/3])[23] Separation, in turn, determines reliability (item separation and person separation reliability) which indicates the degree of confidence that can be placed in the consistency of the estimates (range 0–1; coefficients equivalent to or exceeding 0.80 are considered good which corresponds to a separation index of 2.0).[25] However, person separation reliability depends on both the average SE of the person measure estimate and the standard deviation (SD) of the person measure distribution. Hence, separation reliability is sample-dependent. Given this, we used item separation reliability as an indicator of the instrument's reliability instead of person separation reliability.

To assess the ability of the instrument to appropriately target the population under consideration, we visually inspected the Wright person-item map. A well-targeted instrument should include a set of items that span the full range of person estimates. Targeting and reliability (or measurement precision) are related concepts. Reliability is not uniform across the entire range of test scores. Scores at the edges of the test's range, for example, generally have more error associated with them than scores closer to the middle of the range. Therefore, a Fisher information function is a better method to illustrate targeting and this function defines the upper bound of measurement precision expected by the Rasch model given the observed responses.[19,28] The Fisher information function is an index related to the distribution of item measures but incorporates SE of the estimate for each item. In a well-targeted instrument this would indicate the highest level of information at the mean of the person measures. The Fisher information function was superimposed on the person-item map.

Factor analyses were performed using SPSS software (version 15.0, Chicago, IL). A P value of <0.05 was considered statistically significant.

RESULTS

Assessment of Response Categories

Figure 1a illustrates the CPC of an item in its original 5-category format showing the range of visual function for which each of the five response options were most likely to be chosen. The CPC plots the probability of choosing a given category (on the y-axis) for every place along the measured construct (visual function on the x-axis). Useful categories are those with high probabilities that span a distinct portion of the construct. However, the figure shows that categories 1 ("little difficulty") and 3 ("great deal of difficulty, but can still manage") do not have a range along the scale where they are the most likely categories to be selected and disordering of thresholds is observed. Disordering is also shown by the lack of monotonic increase of the thresholds (from category 1 to 2 and 3 to 4) in Table 3.

A couple of rescoring options were tested, however, a global rescore was found to be the most appropriate solution. All disordered thresholds were resolved (Figure 1b and Table 4) by rescoring that involved reducing the original 5-category scale (0, 1, 2, 3, 4) to a 3-category scale (0, 1, 1, 1, 2) by merging the three middle categories ("a little difficulty", "moderate difficulty", and "great deal of difficulty but can still manage"). The resultant three categories included: 0 "no difficulty", 1 "can manage with difficulty", and 2 "cannot manage". In essence, the end categories were untouched and the intermediate categories were combined in the revised rating scale of the APEDS-VFQ.

Dimensionality

Two items (item 1 "reading small prints in the newspaper/magazines" and item 2 "recognizing small objects,

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for example inserting a thread in the needle") showed misfit, albeit misfit was negligible for item 2 (Table 5). The misfit of item 1 was present both before and after re-organization of the response categories (Tables 5 and 6). PCA of the item measure residuals showed that the variance explained by the principal component was 56%, but the unexplained variance explained by the first contrast was 2.9 eigenvalue units. Thus, the findings of dimensionality were mixed. The finding of item misfit suggests that the item ("reading small prints in the newspaper/magazines") was not in tandem with the rest of the items in the measurement of visual ability. Instead of deleting this item, we tried to understand the source of such a high misfit. So in the next step we examined the fit statistics for the person measures given that item misfit can be sample-dependent.

Figure 2 plots the person measures against the z-transformation (Zstd) infit values for the 351 participants (prior to collapsing categories). Data points for the participants with most visual ability are located at the bottom of the graph and those for participants with least visual ability are located at the top. Data points for participants that are in the shaded regions of the box are those that either exceeded (right box) or were less than (left box) the expected value by 2 SD indicating that these response patterns are inconsistent with the

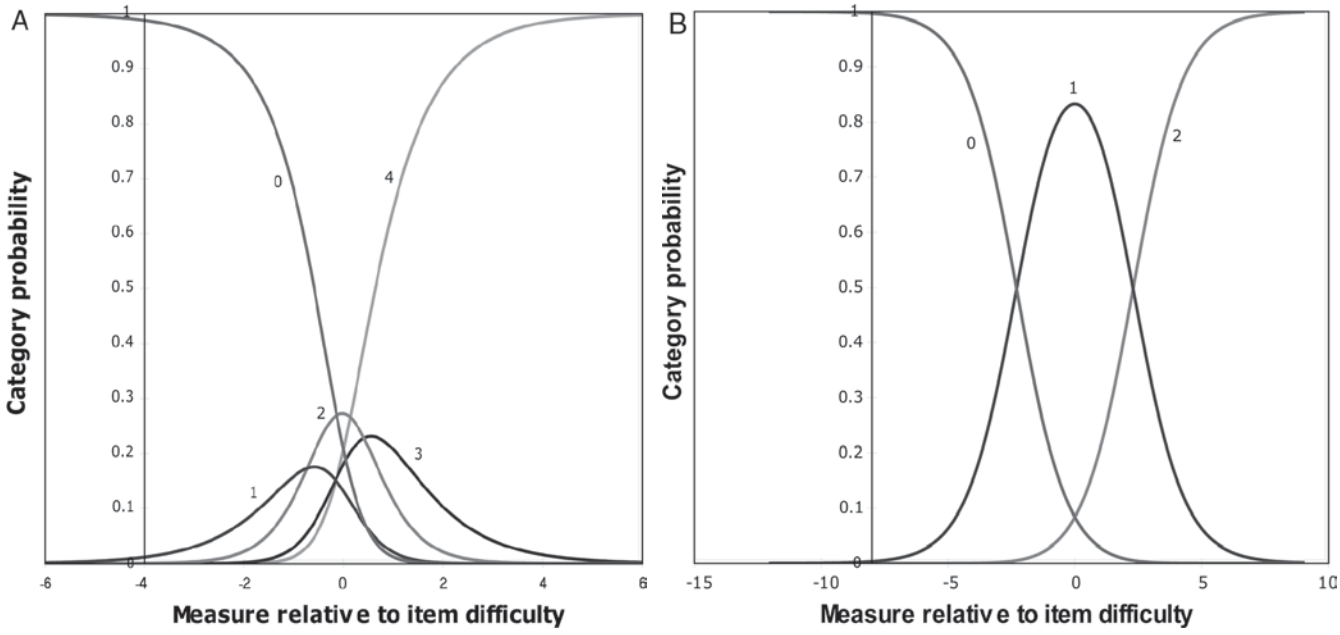


FIGURE 1 (a) Rasch model category probability curves (CPC) for the 16-item Andhra Pradesh Eye Disease Study-Visual Function Questionnaire illustrating the range over which each of the five categories is most likely chosen. Response categories: 0 "no difficulty", 1 "little difficulty", 2 "moderate difficulty", 3 "great deal of difficulty, but can still manage" and 4 "cannot manage". The CPC plots the probability of choosing a given category (on the y-axis) for every place along the measured construct (visual function on the x-axis). Boundaries (or thresholds) occur at points along the scale where the category most likely to be chosen changes from one to the next. For any given point along this scale, the category most likely to be chosen by a participant is shown by the category curve with the highest probability. At no point were categories 1 or 3 the most likely to be chosen, resulting in disordered thresholds. (b) Category probability curves (CPC) following category re-organization of response categories from Figure 1a. Response categories 1, 2 and 3 were combined to form a single category ("can manage with difficulty") with end categories remaining unaltered. The thresholds demonstrated an ordered pattern.

TABLE 3 Response category measures for original rating scale of the 16-item Andhra Pradesh Eye Disease Study-Visual Function Questionnaire (N = 351).

Response category	Observed count (%)	Infit MnSq	Threshold	Category measure
No difficulty (0)	2238 (40)	1.15	None	(−1.51)
Little difficulty (1)	637 (11)	1.06	0.46	−0.58
Moderate difficulty (2)	884 (16)	0.92	−0.73	−0.03
Great deal of difficulty but can still manage (3)	612 (11)	1.12	0.41	0.55
Cannot manage (4)	1212 (22)	0.88	−0.14	(1.59)

Observed count: number (%) of times each response category was used across items and participants. Infit mean square (MnSq): information-weighted fit statistic for each category (expected value 1.0). Threshold: the calibrated measure of the transition from the category below to this category (there is no threshold for category 0, because there is no lower category), thresholds should increase monotonically across the rating scale. Category measure: for each response category 1–3, category measure is the sample-free average functional reserve (functional reserve is the difference between a person's visual ability and ability required by the item). For extreme categories (0 and 4), the category measures are 0.25 logit inside the most extreme functional reserve measures for the sample and are denoted with parentheses as the sample-free functional reserve is infinite for the extreme categories.

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TABLE 4 Response category measures for the revised rating scale of the 16-item Andhra Pradesh Eye Disease Study-Visual Function Questionnaire (N = 351).

Response category	Observed count (%)	Infit MnSq	Threshold	Category measure
No difficulty (0)	2238 (40)	1.12	None	(−2.34)
Can manage with difficulty (1)	2133 (38)	0.92	−1.17	0.00
Cannot manage (2)	1212 (22)	0.96	1.17	(2.34)

Observed count: number (%) of times each response category was used across items and participants. Infit mean square (MnSq): information-weighted fit statistic for each category (expected value 1.0). Threshold: the calibrated measure of the transition from the category below to this category (there is no threshold for category 0, because there is no lower category), the thresholds should increase monotonically across the rating scale. Category measure: For response category 1, category measure is the sample-free average functional reserve (functional reserve is the difference between a person's visual ability and ability required by the item). For extreme categories (0 and 2), the category measures are 0.25 logit inside the most extreme functional reserve measures for the sample and are denoted with parentheses as the sample-free functional reserve is infinite for the extreme categories

TABLE 5 The 16-item Andhra Pradesh Eye Disease Study-Visual Function Questionnaire (original 5 categories) with item calibrations and fit statistics before and after deleting misfitting persons (items ordered from least to most difficult).

Prior to deleting misfitting persons					After deleting most-misfitting persons				
Item no.	n (N = 351)	Measure, logits	SE	Infit MnSq	Item no.	n (N = 331)	Measure, logits	SE	Infit MnSq
14	350	1.14	0.07	0.93	14	330	1.29	0.08	0.93
3	351	0.79	0.06	1.24	3	331	0.89	0.06	1.28
10	351	0.65	0.05	1.04	10	331	0.75	0.06	1.05
8	350	0.57	0.05	0.75	8	330	0.62	0.06	0.78
11	347	0.56	0.05	1.22	11	327	0.60	0.06	1.28
16	350	0.39	0.05	0.77	16	330	0.46	0.05	0.80
12	351	0.31	0.05	0.98	12	331	0.43	0.05	0.91
9	351	0.18	0.05	0.73	9	331	0.23	0.05	0.71
7	351	0.08	0.05	1.30	7	331	0.13	0.05	1.29
15	351	0.06	0.05	0.70	15	331	0.12	0.05	0.69
6	350	0.00	0.04	1.08	13	331	0.04	0.05	0.74
13	351	−0.03	0.04	0.78	6	331	0.02	0.05	1.10
5	347	−0.90	0.05	0.91	5	327	−1.04	0.05	0.91
1	331	−1.07	0.05	**1.54**	1	311	−1.31	0.06	**1.50**
2	350	−1.36	0.06	**1.31**	4	331	−1.57	0.07	1.30
4	351	−1.36	0.06	1.27	2	330	−1.65	0.07	1.29

Measure (logits): estimate for the item difficulty in logits, positive values indicate less difficult items and negative values indicate more difficult items.
SE: standard error of measurement in estimation of item difficulty in logits.
Infit mean square (MnSq): values in bold represent misfitting items (using criterion 0.7–1.3), with item 2 being borderline misfit.

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expectations of the model. A total of 9% of participants had infit Zstd values that exceeded the expected values by more than 2 SD (compared with 2.5% of participants that would occur by chance alone), and 8% of participants had infit Zstd values that were below the expected values by more than 2 SD. However, 6% of participants were considered to be most-misfitting the model (infit Zstd above 2.5) indicating that the response patterns of these participants to the 16 APEDS-VFQ items were inconsistent with the expectations of the model. We did not, however, find anything in their chart histories for such behavior. Nonetheless, in order to ascertain the influence of misfitting participants on item fit statistics, we used the original data set with all 351 participants. Subsequently, we censured participants with infit Zstd > 2.5 and re-ran the Rasch analysis (Table 5). However, this did not help fix the item misfit and item 1 continued to misfit (Table 5). At this stage we decided to retain item 1 as it represents an important activity in daily life (it was applicable to 94% of the sample), and is among the chief complaints among people presenting to low vision clinics. Nonetheless, we investigated DIF (see below) prior to making a final decision on item deletion.

Results of DIF analysis revealed that uniform DIF occurred for groups of misfitting persons and non-uniform DIF varied across infit groups. Notable uniform DIF occurred for item 1 ("reading small prints in the newspaper/magazines").

Notable DIF was present only in the visual acuity group. Those with visual acuity better than 20/200 rated item 1 as relatively easier by 1.24 logits compared to other items. DIF by visual acuity suggested that our sample was heterogeneous (ie we had participants with severe impairment due to field loss but normal visual acuity, mixed with those whose main problem was loss of visual acuity). Therefore, as described in the methods, we performed exploratory factor analysis on four sets of person measures plus logMAR visual acuity and found eigenvalues of 2.4 for the first component and 1.2 for the second. All the resolution and contrast sensitivity items

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TABLE 6 The 16-item Andhra Pradesh Eye Disease Study-Visual Function Questionnaire with item calibrations and fit statistics after collapsing categories (ordered by item difficulty from least to most difficult).

Item no.	n (N = 351)	Measure, logits	SE	Infit MnSq
14	350	2.31	0.14	0.85
3	351	1.61	0.11	1.22
10	351	1.30	0.11	1.00
11	347	1.13	0.11	1.21
8	350	0.83	0.10	0.83
16	350	0.73	0.10	0.76
12	351	0.67	0.10	1.03
7	351	0.30	0.10	1.19
9	351	0.24	0.09	0.76
15	351	0.17	0.09	0.78
6	350	0.16	0.09	1.03
13	351	–0.01	0.09	0.81
5	347	–1.73	0.10	0.90
1	331	–2.18	0.11	**1.43**
2	350	–2.75	0.11	1.27
4	351	–2.79	0.11	1.21

Measure (logits): estimate for item difficulty in logits, positive values indicate less difficult items and negative values indicate more difficult items.
SE: standard error of measurement in estimation of item difficulty in logits.
Infit mean square (MnSq): the value in bold represents a misfitting item.

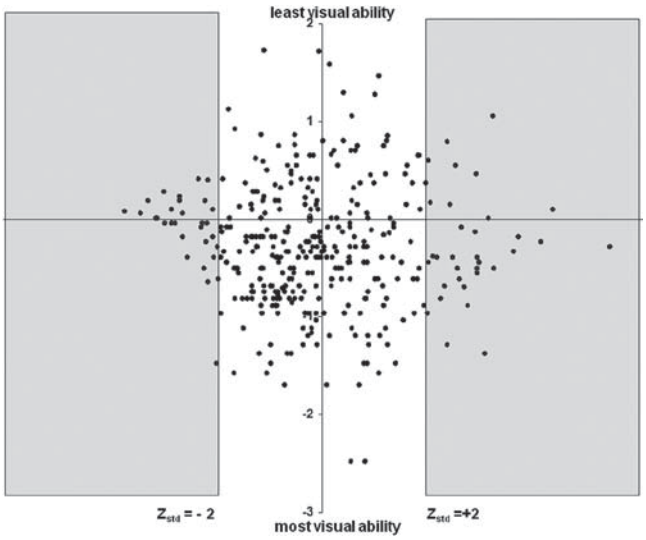


FIGURE 2 Person measures of visual ability estimated from difficulty ratings of 16 items of the Andhra Pradesh Eye Disease Study-Visual Function Questionnaire plotted on the ordinate against the infit mean squares transformed to a standard normal deviate (z-score) on the abscissa. Data points for participants with the most visual ability are located at the bottom of the graph and those for participants with least visual ability at the top. Data points within the boxes represent those participants whose data points lie outside the ± 2 standard deviation (SD) limits around the expected value of the transformed infit mean square. The expected value of the infit mean square is 0 and 95% of values are expected to fall within ± 2SD of the expected value.

loaded heavily, along with logMAR visual acuity, one to one factor, and the peripheral vision and items dependent on illumination levels loaded onto another factor. The first principal component accounted for 72% of variance. These results indicate that the difficulty ratings are dominated by a single variable, visual ability. However, the single most-misfitting item perhaps suggests that there is a two factor composite latent variable in the underlying construct of visual ability. Given this, removing the most-misfitting item would simply bias the construct towards one of the component variables. Thus, we decided to retain the single most-misfitting item (item 1).

TABLE 7 Reliability indices of the Andhra Pradesh Eye Disease Study-Functional Vision Questionnaire (APEDS-VFQ): Native version and with revised rating scale.

Parameter	APEDS-VFQ (native version)	APEDS-VFQ (revised rating scale)
Item separation index	13.24	13.57
Item separation reliability	0.99	0.99
Person separation index	2.08	2.00
Person separation reliability	0.81	0.80

Reliability and Targeting

Reliability indices for the APEDS-VFQ before and after the phase of rating scale modification are shown in Table 7. The reliability indices of the revised rating scale APEDS-VFQ presented comparable values to those of the APEDS-VFQ (native version); all the item and person reliabilities were ≥0.80. The items of both versions were distributed into about 18 difficulty strata.

Figure 3 shows the person-item map (with Fisher information function superimposed) of participant ability and item difficulty of the APEDS-VFQ in the same logit units. It can be seen from the map that the items are not very well targeted to the participant sample. There are very few items (25%) that target participants with the most visual ability and there are large gaps between items.

Criterion Validity

There was a fair and statistically significant relationship ($r = 0.28$, $P < 0.0001$) between binocular visual acuity and the Rasch-scaled APEDS-VFQ score (Figure 4).

Conversion of Raw- to Rasch-scaled Scores

We recommend that users of the revised version of the APEDS-VFQ perform Rasch analysis on their own data. Given that Rasch analysis is considered tedious, and therefore, as stated in our objectives, we have provided ready-to-use Excel spreadsheets for conversion of raw ordinal scores to linear Rasch scores for those who are unfamiliar with Rasch analysis. However, the conversion

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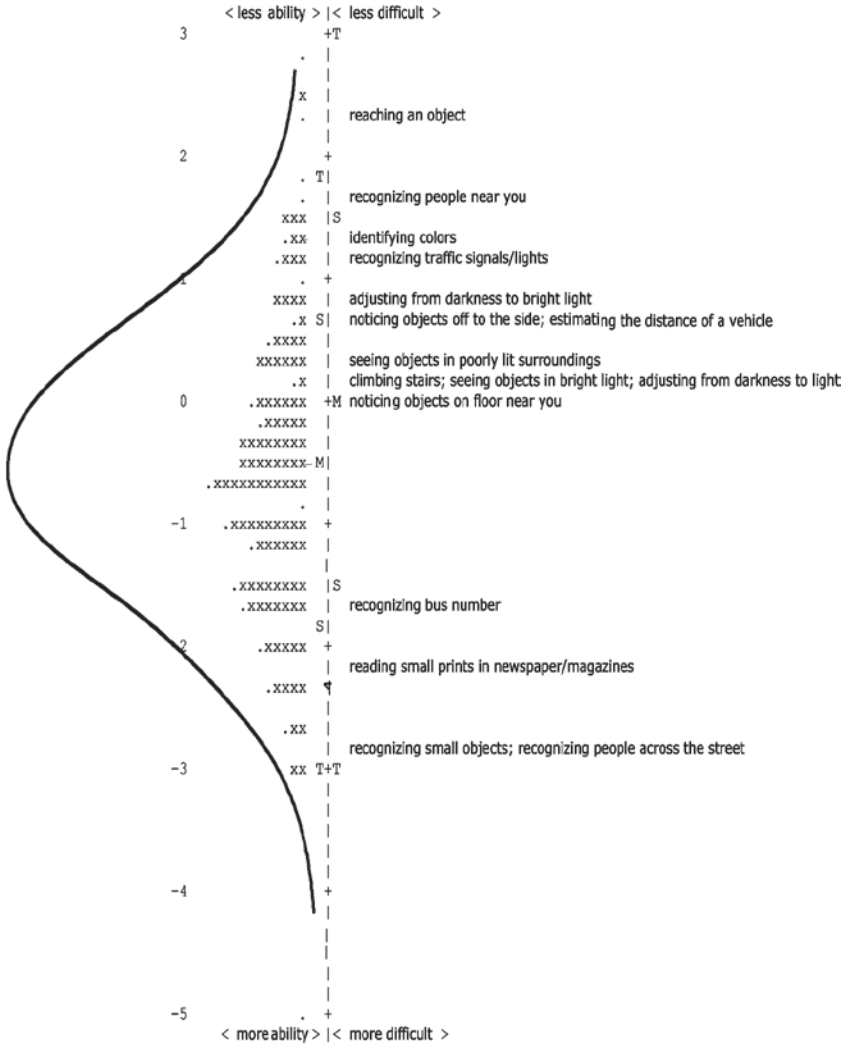


FIGURE 3 Person-item map for the 16-item Andhra Pradesh Eye Disease Study Visual Function Questionnaire in low vision patients. The vertical line represents the measure of visual ability, in logit units. Participants (represented by "x") are located on the left hand side of the map and those with higher visual ability located toward the top. Items are located on the right of the dashed line with less difficult-to-perform items also located toward the top of the map. Full descriptions of items are provided in Table 1. Each "x" represents three participants and each "." represents 1–3 participants. M, mean; S 1 standard deviation (SD) from mean; T, 2SD from the mean. Overlaid is the Fisher information function (black curve).

will be applicable only when the new data show a similar fit to the Rasch model. The Excel spreadsheets can either be downloaded directly from the journal's website or can be obtained by contacting the corresponding author.

Alternatively, the user can refer to the table of category measures, structure calibrations and item measures (Tables 4 and 6) to anchor the values to our calibrations and use Ministeps, which is a free version of Winsteps (Winsteps.com, Beaverton, Oregon), to estimate person measures.

DISCUSSION

The results of the present study are in accordance with similar such studies using other VFQs which concluded that visual ability is a composite variable and can be segregated into two independent dimensions.[11–14] It is probably the choice of the items in a given VFQ that can bias the estimates of visual ability toward one or the other dimension.

However, unidimensionality is one of the requirements of the Rasch model and needs to be satisfied in case items are summed to generate a valid summary score. Despite the finding from the present study of the presence of two dimensions in visual ability, we believe that as is the case with other most widely used VFQs,[12,14] the APEDS-VFQ also functions as if it were unidimensional and, therefore, should be considered as a composite variable. In the present study, reading and contrast sensitivity items loaded heavily on one factor, and peripheral vision and items dependent on illumination loaded on another. Reading and contrast sensitivity, and peripheral vision and illumination are not measures of independent factors; rather they are measures that incorporate both factors. One dimension most heavily

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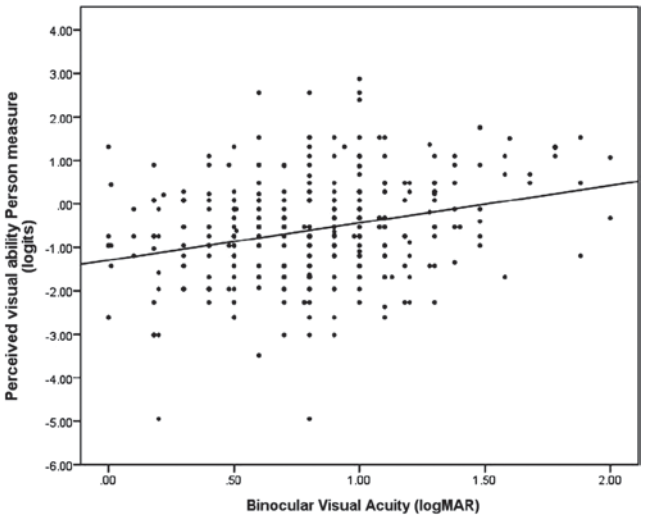


FIGURE 4 Scatterplot of person measure estimates based on participants' responses to 16 items that required difficulty ratings on the Andhra Pradesh Eye Disease Study-Visual Function Questionnaire versus presenting binocular visual acuity. A logarithm of the minimum angle of resolution (logMAR) of 0 corresponds to Snellen acuity of 20/20 and a logMAR of 1 corresponds to 20/200. Negative values of person measures indicate higher visual ability and positive values lower visual ability. The regression line (solid line) has a slope of 0.86 and an intercept of −1.29. The Pearson correlation coefficient is 0.29.

influences responses to items that describe reading and contrast sensitivity items and both require high visual resolution. This dimension is related to seeing detail that is disrupted by loss of visual acuity. The other dimension most heavily influences responses to items that describe peripheral vision and illumination. This dimension is related to size of the seeing area and level of lighting that is disrupted by contrast sensitivity and visual field loss. These two factors together constitute the visual ability variable and explain 72% of the response variance. The low correlation between visual acuity and person measures further supports the two dimensions in visual ability. The two factors mentioned above probably describe the properties of the participant's visual ability rather than the visual impairment. Furthermore, the functional difficulties of low vision patients are varied and the items included in a VFQ must be as comprehensive as possible so as to include the entire range of activities.

In the case of the APEDS-VFQ used in the present study, we found a single most-misfitting item ("reading small prints in the newspaper/magazines"). The large MnSq value for this item suggests that there is high variability in the sample in the functional requirements of the activity. We speculate that such high variability is perhaps related to the double-barreled nature of this item, in that it relates to reading small prints in the newspaper and/or magazines; print size and contrast vary among newspapers and across newspapers and magazines. Consequently, the visual demand could be expected to be different for reading newspapers versus magazines. For example, a participant may have had "mild difficulty" with reading small prints in newspapers but may have had a "great deal of difficulty" with reading such print in magazines. Despite this the participant was required to respond to this item as though it is a single activity. Normally, researchers would delete such a misfitting item so as to optimize the measurement properties of an instrument,[13,29,30] but we did not pursue this. Care needs to be exercised while identifying misfitting items, as the ability of a single fit statistic to correctly identify misfit across a range of assessment scenarios and instruments has been questioned in the Rasch literature.[31] It is therefore suggested that the removal of these items should be assessed against impact on clinical utility.[29] Although the impact was not assessed in the present study (as it was not the intent of this study), our decision to retain the single most-misfitting item appears logical given that it improves content validity as "reading small prints in newspapers, books or magazines" is among the chief presenting complaints of people to a low vision clinic. This is evidenced by the fact that the item was applicable to 94% of our participants. Furthermore, there was no substantial worsening or improvement of the item or person separation reliability indices, further supporting our decision to retain this item.

Rather than delete misfitting items, their presence should suggest the existence of a problem which requires further investigation (such as DIF analysis). Furthermore, item misfit can be sample-dependent, so we investigated fit statistics for the person measures. Sometimes removing grossly misfitting persons could fix the item misfits. However, the number of most-misfitting persons was probably too small to alter the item fit statistics and the single item continued to misfit. In our DIF analysis, we found the same most-misfitting item to demonstrate notable DIF by visual acuity suggesting again that it is a potential candidate for deletion, as the presence of DIF violates one of the requirements of unidimensionality of the model. Nonetheless, the existence of DIF in itself is not an indication of a problem with the item. However, just as judgment is required to determine cut-offs to flag DIF items, judgment is required to determine if the item is essential to the measurement process. Instruments are often a mixture of items that don't measure the same thing in the same way. Items that exhibit performance differences on the rest of the instrument will be flagged for DIF but may be essential to the construct of interest. Furthermore, DIF may occur when an item measures dimensions other than the dimension measured by the other items. In the case of the APEDS-VFQ, examination of the items and the pattern of DIF indicated that slight multidimensionality was due to two dimensions of interest to the researcher (as mentioned earlier).

Dimensionality *per se* is a bad property of an instrument if the construct of interest is a composite variable (such as the visual ability variable in most VFQs including the APEDS-VFQ). In contrast, if the dimensionality represents subgroups of participants that behave differently, then the construct is not a composite, but rather is confounded, which is undesirable. Given that the

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underlying construct is a composite variable, removing a misfitting item(s) would simply bias the construct towards one of the component variables. This further supported our decision against deleting the single most-misfitting item and we thus retained all the original 16 items in the instrument.

Among the desirable qualities of a good measure is adequate reliability. The high item-separation reliability found for the APEDS-VFQ in the present and earlier study suggest that the items on the APEDS-VFQ are well separated relative to errors of their location on the scale, thereby defining a clear scale and that great confidence can be placed in the replicability of the item placement across future samples. Furthermore, it also suggests a high probability that items with high difficulty estimates are really more difficult than items with lower difficulty estimates.[32] Thus, it would be safe to conclude that the APEDS-VFQ is a precise measure of visual ability in the visually impaired in India. Similar high item-separation reliability has been reported for the Activity Inventory developed by Massof.[12,13]

A well-targeted instrument is a logical prerequisite when measurement and evaluation of improvements in patient outcomes are desired. In the present study, items mostly targeted participants with lower visual ability and gaps remained for those with relatively higher visual ability. More items (difficult) may be needed at the lower end of the scale to distinguish among levels of those with higher visual ability.

Inspection of the rating scale format for the APEDS-VFQ revealed issues regarding the ordering of response categories indicating that the response scale does not work as intended. Disordering of thresholds was identified but a reduction in number of response categories from five to three resolved any disordering. In the earlier Rasch analysis of the APEDS-VFQ, thresholds were also disordered and combining two middle categories ("moderate difficulty" and "great deal of difficulty but can still manage") was suggested. However, in the present study there was a need to combine all three middle categories (including "little difficulty"), indicating that participants were unable to reliably distinguish between what is meant by the terms "a little", "moderate" and "a great deal", which could be replaced by a single category (ie "some difficulty") for optimal functioning of the response categories. Despite the reduction in the response options from five to three, the revised APEDS-VFQ scale showed good item and person-separation reliability supporting this revision. Nonetheless, the new rating scale needs to be validated in future studies prior to being recommended for use.

This study builds on previous Rasch analysis of the APEDS-VFQ. In the present study, we specifically assessed the dimensionality of the instrument. Like other VFQs, we found the latent variable (visual ability) of the APEDS-VFQ to be a two-dimension composite variable. The high reliability (for both items and persons) of the APEDS-VFQ (with the new rating scale) shown in this study further supports the good measurement properties demonstrated in the earlier Rasch analysis. However, further studies are needed. They should include an analysis of the actual performance of the new response categories, because collapsing five categories to three is not the same as presenting three categories to participants.

ACKNOWLEDGMENTS

The authors wish to thank all the participants who volunteered to participate in the study. The authors also wish to thank the anonymous reviewers who helped improve the manuscript significantly.

***Funding/Support*:** This study was supported in part by the Hyderabad Eye Research Foundation.

***Declaration of interest*:** The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the article.

REFERENCES

[1] Dandona R, Dandona L, Naduvilath TJ, et al. Design of a population-based study of visual impairment in India: The Andhra Pradesh Eye Disease Study. *Indian J Ophthalmol* 1997;45(4):251–257.
[2] Nutheti R, Shamanna BR, Krishnaiah S, et al. Perceived visual ability for functional vision performance among persons with low vision in the Indian state of Andhra Pradesh. *Invest Ophthalmol Vis Sci* 2004;45(10):3458–3465.
[3] Mallinson T. Why measurement matters for measuring patient vision outcomes. *Optom Vis Sci* 2007;84(8):675–682.
[4] Tennant A, Pallant JF. Unidimensionality matters. *Rasch Meas Trans* 2006;20:1048–1051.
[5] Tennant A, McKenna SP, Hagell P. Application of Rasch analysis in the development and application of quality of life instruments. *Value Health* 2004;7(Suppl. 1):S22–26.
[6] Conrad KJ, Smith EV, Jr. International conference on objective measurement: applications of Rasch analysis in health care. *Med Care* 2004;42(Suppl. 1):I1–6.
[7] Tesio L. Measuring behaviours and perceptions: Rasch analysis as a tool for rehabilitation research. *J Rehabil Med* 2003;35(3):105–115.
[8] Smith EV. Detecting and evaluating the impact of multidimensionality using item fit statistics and principal component analysis of residuals. *J Appl Meas* 2002;3:205–231.
[9] Angoff WH. Use of difficulty and discrimination indices for detecting item bias. In: Berk RA, ed. *Handbook of methods for detecting test bias*. Baltimore, MD: Johns Hopkins University Press, 1982.
[10] Holland PW, Wainer H. *Differential item functioning*. Hillsdale: Lawrence Erlbaum Associates, 1993.
[11] Massof RW, Ahmadian L, Grover LL, et al. The Activity Inventory: an adaptive visual function questionnaire. *Optom Vis Sci* 2007;84(8):763–774.
[12] Massof RW, Hsu CT, Baker FH, et al. Visual disability variables. II: the difficulty of tasks for a sample of low-vision patients. *Arch Phys Med Rehabil* 2005;86(5):954–967.
[13] Massof RW, Hsu CT, Baker FH, et al. Visual disability variables. I: the importance and difficulty of activity goals for a sample of low-vision patients. *Arch Phys Med Rehabil* 2005;86(5):946–953.
[14] Massof RW. An interval-scaled scoring algorithm for visual function questionnaires. *Optom Vis Sci* 2007;84(8):689–704.

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[15] Marella M, Gothwal VK, Pesudovs K, Lamoureux E. Validation of the visual disability questionnaire (VDQ) in India. *Optom Vis Sci* 2009;86(7):E826–835.
[16] Bailey IL, Lovie JE. New design principles for visual acuity letter charts. *Am J Optom Physiol Opt* 1976;53: 740–745.
[17] Andrich DA. A rating scale formulation for ordered response categories. *Psychometrika* 1978;43:561–573.
[18] Massof RW. The measurement of vision disability. *Optom Vis Sci* 2002;79(8):516–552.
[19] Massof RW. Understanding Rasch and item response theory models: applications to the estimation and validation of interval latent trait measures from responses to rating scale questionnaires. *Ophthalmic Epidemiol* 2011;18(1):1–19.
[20] Linacre JM. WINSTEPS Rasch measurement computer program. Chicago: Winsteps.com, 2008.
[21] Gothwal VK, Wright TA, Lamoureux EL, Pesudovs K. Rasch analysis of the quality of life and vision function questionnaire. *Optom Vis Sci* 2009;86(7):E836–844.
[22] Gothwal VK, Wright TA, Lamoureux EL, Pesudovs K. Activities of Daily Vision Scale: what do the subscales measure? *Invest Ophthalmol Vis Sci* 2010;51(2):694–700.
[23] Wright BD, Masters GN. Rating scale analysis. Chicago: MESA Press, 1982.
[24] Smith AB, Rush R, Fallowfield LJ, et al. Rasch fit statistics and sample size considerations for polytomous data. *BMC Med Res Methodol* 2008;8:33.
[25] Bond TG, Fox CM. Applying the Rasch model: fundamental measurement in the human sciences. London: Lawrence Erlbaum Associates, 2001.
[26] Bachman LF. Statistical analyses for language assessment. Cambridge: Cambridge University Press, 2004.
[27] Johanson GA, Alsmadi A. Differential person functioning. In: *Annual meeting of the American Educational Research Association*. San Diego, CA: 1998.
[28] Mislevy RJ, Sheehan KM. Information matrices in latent variable models. *J Edu Stat* 1989;14:335–350.
[29] Smith AB, Wright P, Selby PJ, Velikova G. A Rasch and factor analysis of the Functional Assessment of Cancer Therapy-General (FACT-G). *Health Qual Life Outcomes* 2007;5:19.
[30] Lamoureux EL, Pallant JF, Pesudovs K, et al. The Impact of Vision Impairment Questionnaire: an evaluation of its measurement properties using Rasch analysis. *Invest Ophthalmol Vis Sci* 2006;47(11):4732–4741.
[31] Karabatsos G. A critique of Rasch residual fit statistics. *J Appl Meas* 2000;1(2):152–176.
[32] Wright BD, Stone MH. Best test design. Rasch measurement. Chicago: MESA Press, 1979.

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